

LM-318 LASER DUST MONITOR

USER MANUAL

(SPM-UM-023)



AADHAV INTECH
#60, 3rd Street, Kasi Estate,
West Jafferkanpet, Chennai – 600083
Contact No: 044 42041270/9629466446
Tamil Nadu, India

CONTENTS

1.	INTRODUCTION	3
2.	GENERAL SAFETY& PRECAUTIONS	5
3.	LM-318 INSTALLATION REQUIREMENTS	6
4.	INSTALLATION	9
5.	MONITOR UNIT DISPLAY FEATURES	11
6.	SPECIFICATIONS	12
7.	DRAWING	14
7.1.	MONITOR UNIT MECHANICAL DRAWING.	14
7.2.	TRANSMITTER UNIT MECHANICAL DRAWING	15
7.3.	TRANSMITTER UNIT CIRCUIT COMPONENT LAYOUT	16
7.4.	RECEIVER UNIT MECHANICAL DRAWING	17
7.5.	RECEIVER UNIT CIRCUIT COMPONENT LAYOUT	18
8.	CONNECTION DIAGRAM	19
9.	MODBUS CONNECTION	20
10.	MAINTENANCE	21
10.1.	DOS & DONTs	21
10.2.	PERIODIC MAINTENANCE	21
10.3.	TROUBLESHOOTING	22

1. INTRODUCTION

LM-318 is an in-situ, non-contact, cross-stack, optical laser dust monitor which measures dust concentration by measuring opacity of the transmission medium. These monitors are designed for measurement across stacks, pipes and ducts with diameters from 0.5 m to 10 m. LM-318 uses a transmitter/receiver configuration to measure the opacity/dust concentration along the line of sight. LM-318 uses the principle of light attenuation due to scattering and/or absorption in which a laser light transmitted by the transmitter is scattered by dust particles. The incident light is then collected by a solid-state optical detector for dust quantification.

Opacity can be explained as follows. If a beam of light with frequency ν , travels through a medium with opacity k_ν , and mass density ρ , both constant, then the intensity will be reduced with distance x according to the formula

$$I(x) = I_0 e^{-k_\nu \rho x}$$

where

x is the distance the light has travelled through the medium

$I(x)$ is the intensity of light remaining at distance x

I_0 is the initial intensity of light, at

For a given medium at a given frequency, the opacity has a numerical value that may range between 0 and infinity, with units of length²/mass.

Opacity in air pollution work refers to the percentage of light blocked instead of the attenuation coefficient (aka extinction coefficient) and varies from 0% light blocked to 100% light blocked.

$$Opacity = 100\% \times \left(1 - \frac{I(x)}{I_0}\right)$$

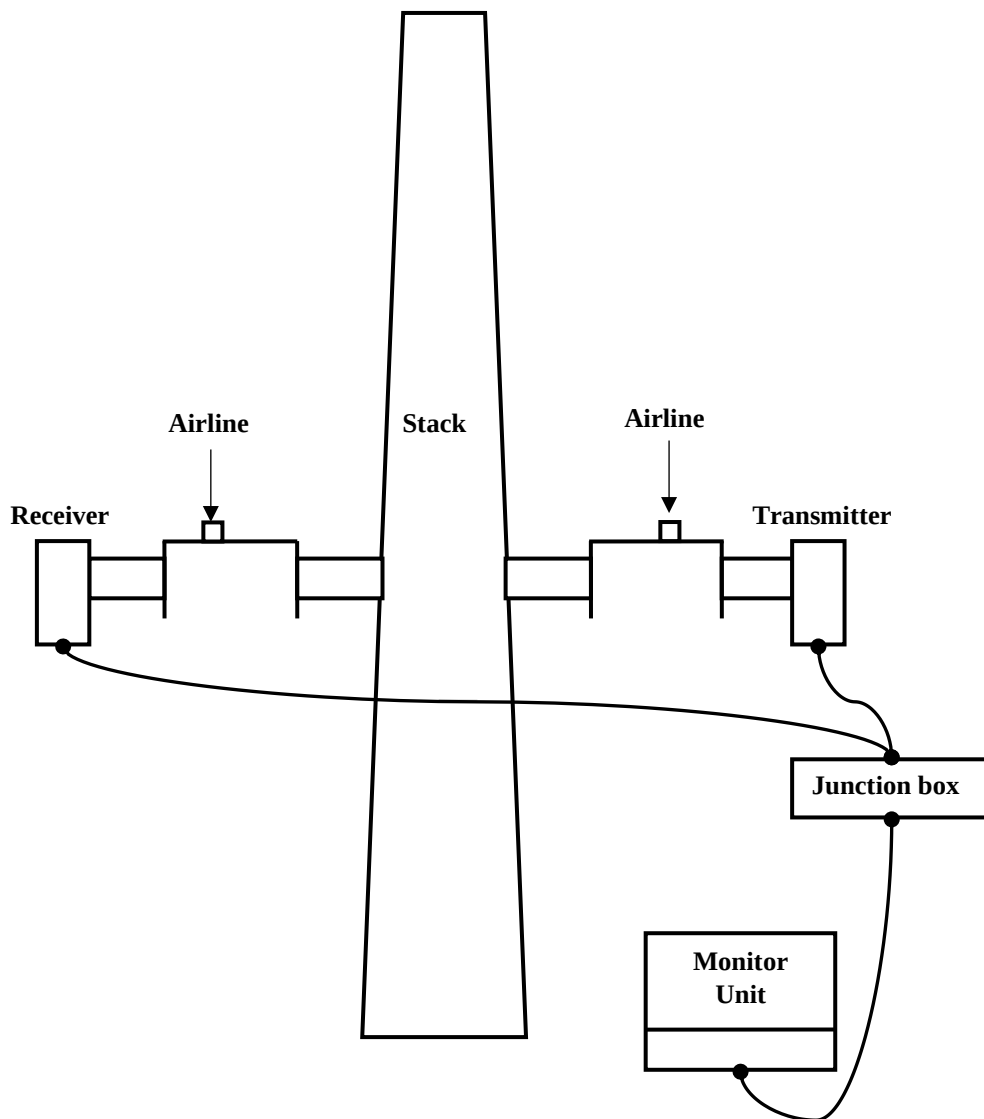
Since LM-318 utilizes a simpler approach for measuring dust concentration, it has no moving parts reducing maintenance costs. Opacity monitor's optimal use is for stack up to 10 m.

LM-318 has the following benefits

- Fast response time
- Reasonable cost
- Process control
- No requirement of sample conditioning

LM-318 include a transmitter, a receiver and a monitor unit. There are no moving parts, so it has minimal maintenance. It has good stability and reliability. Typical applications of dust monitor include power plants, cement factories and tunnels. The alignment of the beam is easy as the beam is narrow.

SCHEMATIC SKETCH OF LASER DUST MONITOR



2. GENERAL SAFETY& PRECAUTIONS

Avoid unsafe acts and conditions. Do not operate or install the instrument before reading this chapter. Installing the LM-318 involves wiring high voltage equipment. Installation must be performed by someone who understands the dangers of, and is qualified to, wire electrical devices. If information or instructions are not clear, DO NOT PROCEED until clarification can be obtained. Follow all safety instructions as given below.

- 2.1. Follow basic safety rules when working with or near high voltage circuits. The LM-318 is a high voltage instrument using 220V/110V. Peripheral devices are also high voltage devices. Keep away from live circuits.
- 2.2. Do not change parts or make adjustments inside the equipment with high voltages on.
- 2.3. Do not service alone.
- 2.4. Do not energize equipment if there is any evidence of water leakage.
- 2.5. All wiring that involves connections to mains power must be performed by a qualified licensed electrician, and must conform to all locally applicable electrical codes.
- 2.6. Do not make connections while power is applied. Turn off power and assure power “Lockout” before installing or servicing to avoid contact with electrically powered circuits. This includes working on devices connected to the relay outputs and auxiliary input systems.
- 2.7. Disconnect external power to LM-318 before connecting or disconnecting components and/or peripheral devices.
- 2.8. Install and use the LM-318 only in accordance with information documented in this manual.

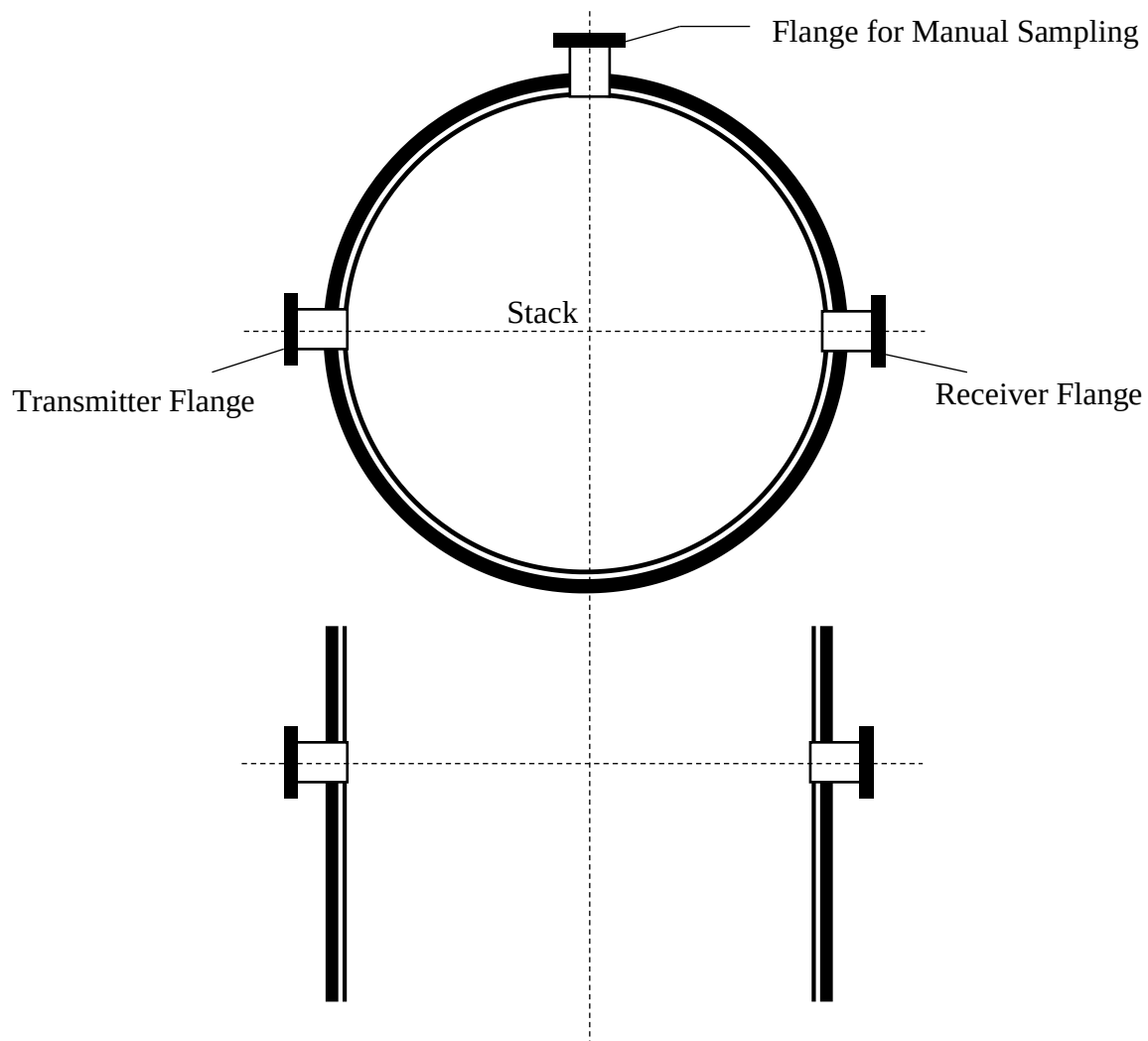
3. LM-318 INSTALLATION REQUIREMENTS

The following utilities are required for the installation of LM-318 dust monitor.

- 3.1. **Operating Conditions:** The installation site should be free from vibrations, dust, heat and electromagnetic generating sources. The instrument display unit should not be exposed to the direct sunlight.
- 3.2. **Power Supply:** The Instrument operates on 220/110 V $\pm$ 10%, 50/60 Hz power supply. UPS power supply is recommended. Two numbers of 5 Amps power socket should be available near to the instrument monitor unit. Always use properly grounded power supply. The sensitivity and stability of the LM-318 will be impaired if it is not grounded. To avoid possible electrical shock or damage to the equipment, connect earth ground to the LM-318 power supply unit.
- 3.3. **Measurement output:** LM-318 provides two analog outputs a 4-20 mA current loop and 0-10 V voltage output which can be connected to control room for viewing the data. Signal cable has to be laid between Display unit and control room to view data at Control Room.
- 3.4. **Sampling Port at Stack:** Sampling ports shall be provided by the user to fit the LM-318 to the stack. The sampling ports are fitted to the stack as shown in the following figure. The sampling ports are fitted exactly 180° to each other on the stack. To ensure laminar flow, sampling ports shall be located at minimum 8 times chimney diameter downstream and 2 times upstream. This will keep the flow away from any disturbance. For a rectangular cross section, the equivalent diameter (D_e) shall be calculated by using following equation to determine up stream, downstream distances.

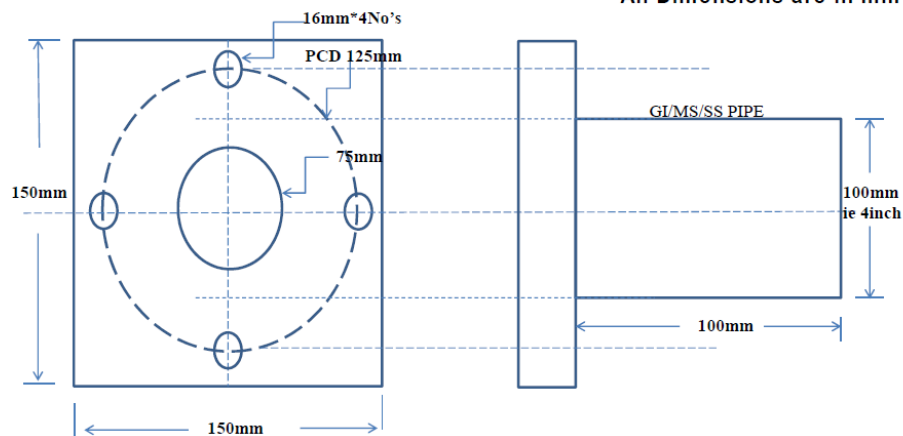
$$D_e = \frac{2LW}{L + W}$$

Where L =Length in m, W = width in m. The stack flanges diagram is also shown in the figure.



STACK FLANGE FOR DUST MONITOR- 2 No's

All Dimensions are in mm



3.5. Other General Requirements:

- 3.5.1. ½” Instrument Air line of greater than 3kg/cm² pressure with isolation valve and air filter regulator is required for dust monitor purging at chimney flanges on both sides.
- 3.5.2. Provide a weather proof junction box with suitable DIN-rail track for connecting eight numbers of wires coming from transmitter and receiver unit and it has to be mounted near transmitter or receiver unit.
- 3.5.3. Eight core signal cable has to be laid from weather proof junction box and display unit fixed in the analyser room.
- 3.5.4. Cable tray should be laid from transmitter, receiver unit to junction box and one more cable tray has to be provided from junction box to display unit).
- 3.5.5. Two numbers of ‘U’ type flanges are required at chimney stack for installation of transmitter unit and receiver unit.

Caution: Warranty is void if failure due to “No air supply” and/or Surge/spike/interrupted power supply.

Note: Instrument Air should be regulated and free from dust, moisture and oil mist.

4. INSTALLATION

The installation environment is different in each application. The gas laser has a specific installation foot and semiconductor lasers as well as receivers, come with the flanges that have elliptic holes for adjustment.

The steps shall be followed during installation of LM-318.

- a) Use 10/11 mm spanner to fit the transmitter unit to the T-U-Flange.
- b) Use 8/9 mm spanner to adjust/align the angle of the transmitter unit.
- c) Use 10/11 mm spanner to fit the receiver unit to another R-U-Flange.
- d) Use adjustable wrench to connect ½ inch to 12 mm pneumatic tube connector to both R-U-Flange and T-U-Flange.
- e) Use a 12 mm to 8 mm pneumatic reducer to connect the transmitter & receiver to their respective U-Flange pneumatic connectors. Use T-joint pneumatic connectors wherever necessary.
- f) Main alignment adjustment should be done from transmitters installation flange.

As the laser beam is very compact, only small holes (10...50 mm dia) to get beam through are needed. Normally there will be a small under pressure inside the stack, and outside air automatically keeps the holes clean. In case there is overpressure in the stack, but source and receiver can be flange-mounted. Then, using the instrument-air purge system to slightly overcome stack pressure keeps the instrument clean and functional.

The mounting system obviously varies from case to case. Basically, there are three different mounting systems:

- a) Mounting in the stack itself by L-type supports.
- b) Mounting on the support structures (or on the floor/roof) around the stack. In this case possible temperature related movement of the stack may require elliptical holes in the stack.
- c) Mounting on the flanges welded in the stack. This case provides a solution, when there is overpressure in the stack and gas escape from small holes is not allowed. A specific problem appears when the dust content is very high. Then you might need to shorten

the laser path by horizontal tube inside the stack. As a rule of thumb, 1 m distance is enough for concentrations 1-5g/m³. Higher concentrations need shorter path.

There are other environmental factors that can influence the measurement. Here is a check list of the factors that can cause trouble for measuring dust content:

Factor 1: Trying to measure too small concentrations.

Factor 2: Using too long distance for very high concentrations.

Factor 3: Orientation and vibration problems when using long measuring distances.

Factor 4: Interfering other substances, than the measured dust.

Factor 5: Other light sources (sun, lighting) affecting the receiver.

Factor 6: Dust or dew accumulating to optical surfaces of the laser or receiver.

Factors 1 and 2 relates to measuring angle and inherent intensity fluctuation in LM-318, as discussed in connection with mass density estimation. The mass formula can be used to estimate the mass concentration range, based on particle properties and path length.

Factor 3: Refers to allowed beam direction changes that can point the beam outside the detector. For the semiconductor model this is 25...35m. For critical cases, larger area available. Consult the factory.

Factor 4: The selected wavelength of the laser is such, that most common 3-atomic gases SO₂, CO₂, H₂O, will not affect the measurements. However, when water exists as liquid droplets or fog, it will be “seen” by the laser as particles. Applications of this kind, such as involving wet scrubbers, should be checked carefully for this kind of trouble.

Factor 5: Sun or other artificial light sources, can add to the laser and thus falsify the results. For modulated laser beam this is not a problem, as the receiver is tuned to modulation frequency.

Factor 6: This is a major problem in competing units, but not normally in laser installations, that are installed remote to stack. Both laser and receiver should be provided with dry, clean instrument air to keep slight overpressure inside the units.

5. MONITOR UNIT DISPLAY FEATURES



The above display shows mg/m^3 –reading of the dust concentration.

The monitor unit features a 4.3-inch TFT screen display and three photo LEDs (Power, Alarm, Status) as user interface. The TFT screen can be used for a range of functions such as set range, calibrate, set date/time, set alarm limit, calculate range, etc. The functions are password protected so as to protect the instrument from any unauthorized use.

- **Power indicator LED -RED:**

Whenever the monitor is powered, this LED is switched ON.

- **Status indicator LED-YELLOW:**

During normal operation, the status indicator LED continuously blinks ON/OFF with a time-period of approx. 1 second.

6. SPECIFICATIONS

6.1. GENERAL SPECIFICATIONS

Unit of Measure	mg/m ³
Range	0...2000 mg/m ³ (user configurable)
Settling time	1 s
Accuracy	± 1 % full scale
Resolution	1 mg/m ³ , 0.1 mg/m ³ (on request)
Analog Output formats	Current 4 – 20 mA, 500Ω max. Voltage 0 – 10 VDC, 1kΩ output impedance.
Digital Output formats	Modbus RTU over RS485
Relay outputs	Alarm buzzer
Power requirement	220VAC 14VA <i>Optional:</i> 110 VAC 14VA
Operating temperature	0°C to +70°C (32°F to 158°F)
Humidity	10% to 90% RH

6.2. TRANSMITTER UNIT (T)

Laser type	Semiconductor laser class II
Optical Power	0.9 mW nominal
Wavelength	715 nm, visible light
Power stability	± 1% warm-up 5 minutes
Power supply	± 15V from monitor unit
Operating temperature	-20°C to +60°C (-4°F to 140°F)

6.3. RECEIVER UNIT (R)

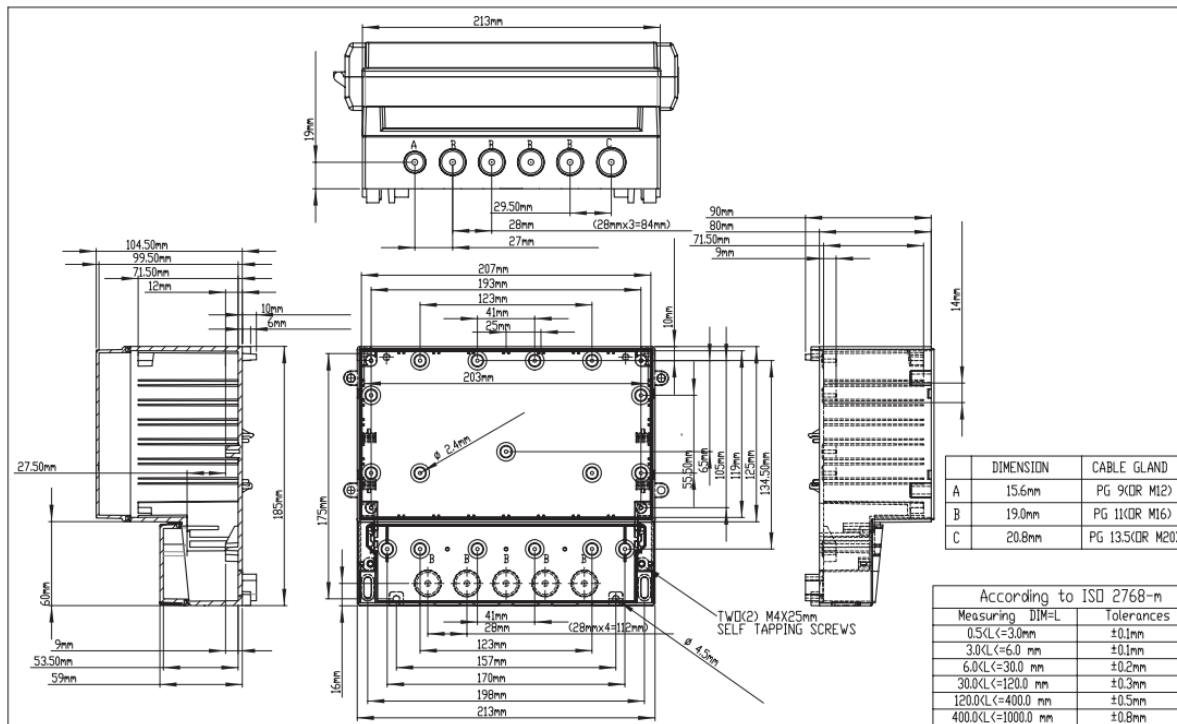
Detection	Optically matched semiconductor Detector with Ø 50mm glass lens.
Power supply	±15V, from monitor unit
Operating temperature	-20°C to +60°C (-4°F to 175°F)

6.4. MONITOR UNIT (M)

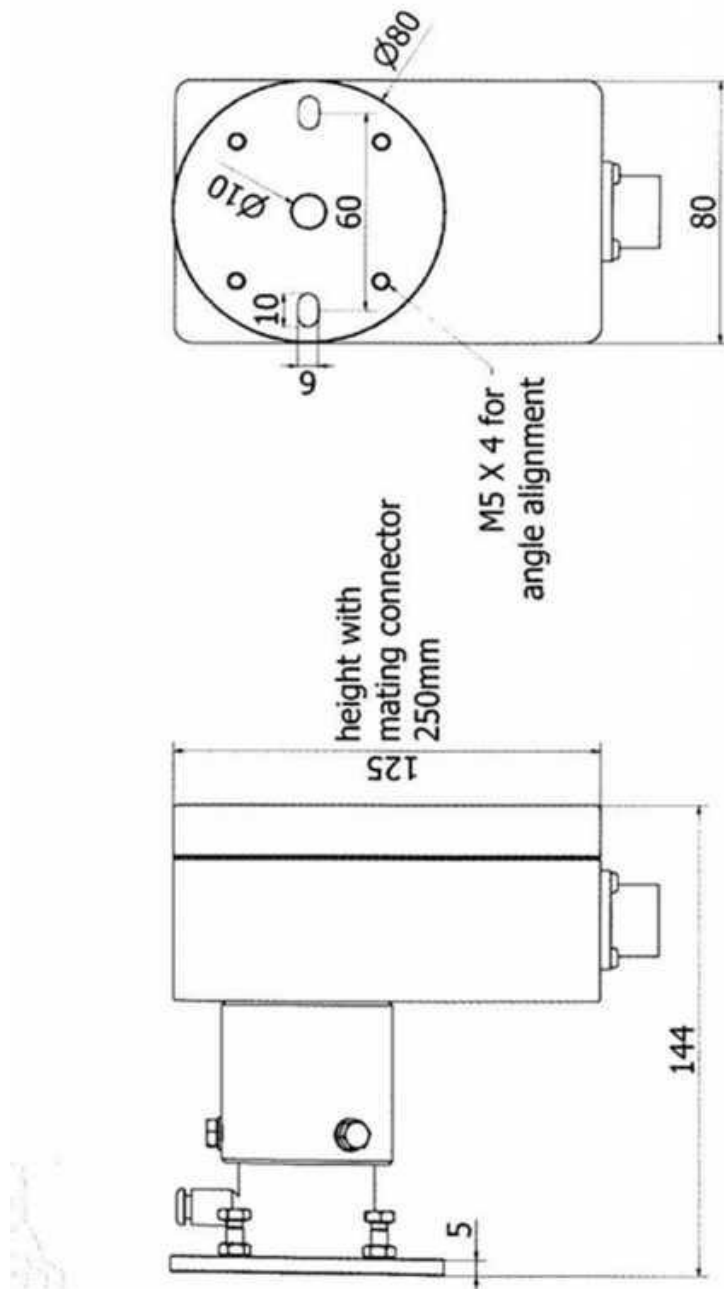
Power requirement (from SMPS unit)	+ 15 VDC @ 2 A ± 0.5 V - 15 VDC @ 1 A ± 0.5 V +5 VDC @ 5 A ± 0.1 V +24 VDC @ 1A ± 0.5 V
Operating temperature	0°C to +70°C (32°F to 158°F)
Dimension in mm	213 * 185 * 104.5
Material	ABS plastic UL-94HB
Colour	Light Grey
Cap	Transparent
Impermeability	Yes
Degree of protection	IP65

7. DRAWING

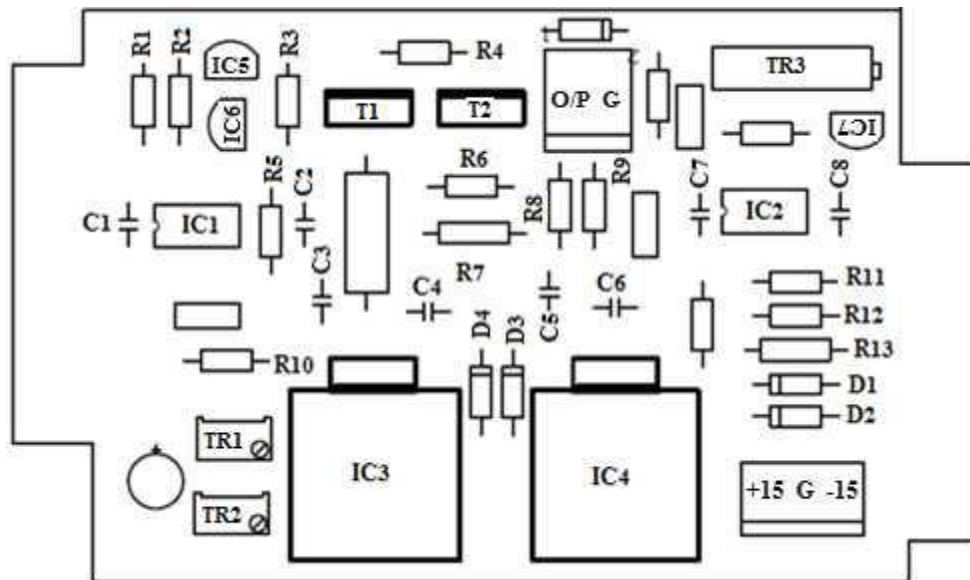
7.1. MONITOR UNIT MECHANICAL DRAWING



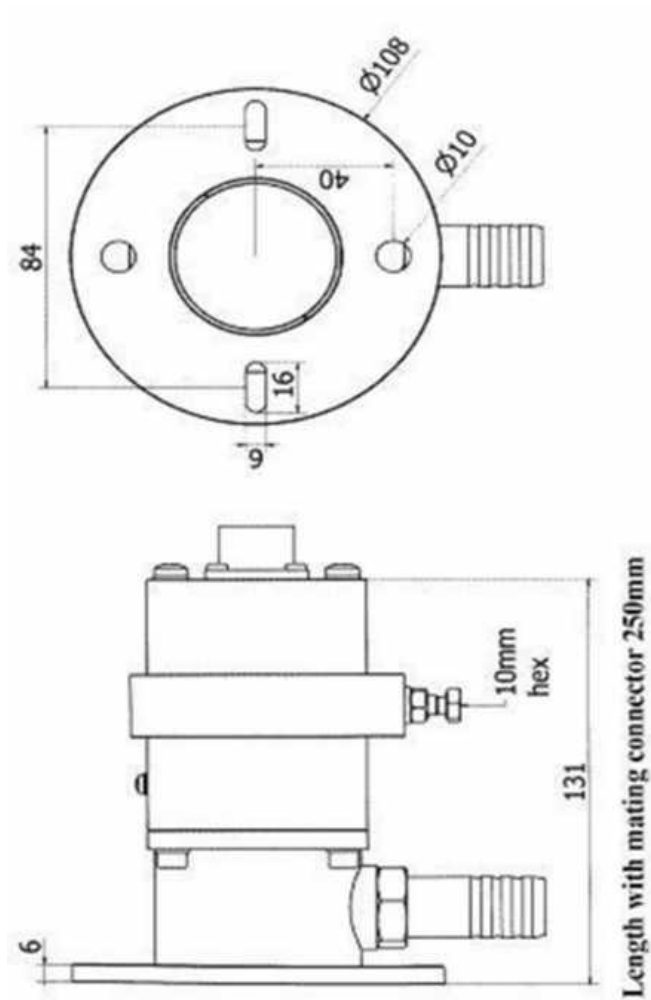
7.2. TRANSMITTER UNIT MECHANICAL DRAWING



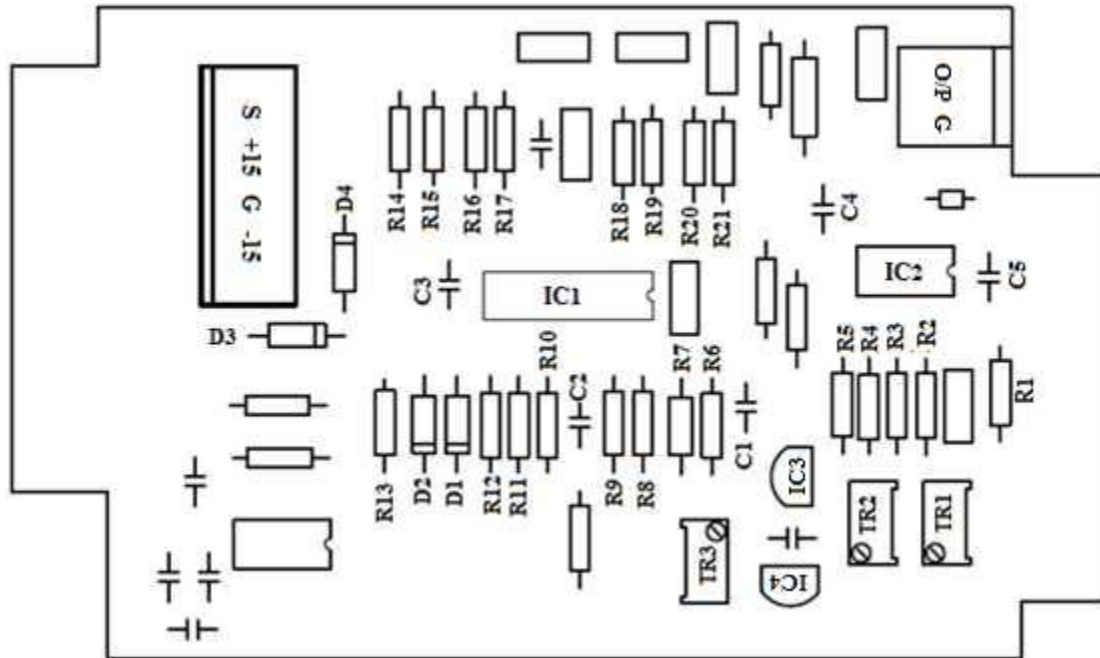
7.3. TRANSMITTER UNIT CIRCUIT COMPONENT LAYOUT



7.4. RECEIVER UNIT MECHANICAL DRAWING



7.5. RECEIVER UNIT CIRCUIT COMPONENT LAYOUT



8. CONNECTION DIAGRAM

1	2	3	4	5	6	7	8	9	10	11	12
SMPS						LASERMOD				AUX	
+15V	-15V	+5 V	G	24V+	24V-	+15V	-15V	SIG	GND	SIG AUX	GND

13	14	15	16	17	18	19	20
VOLTAGE OUT (0-10V)		CURRENT OUT (4-20mA)		BUZZER		RS485	
VOUT	GND	I+	I-	P	NO	A+	B-

9. MODBUS CONNECTION

LM-318 provides MODBUS-RTU communication port through standard RS485 port at a fixed baud rate of 9600 with 8-N-1 bit format. The dust concentration values are available in the Holding registers starting from address 40001 and can be read using Function Code 03 with standard 16-bit-CRC. The MODBUS registers and their addresses are as follows.

S.No	Register Address	Measured Parameter	Unit	Format
1.	40001	Dust Concentration	mg/m ³	Swapped Float

10. MAINTENANCE

10.1. DOS AND DONTs (LM-318 SPECIFIC)

- 10.1.1. Always use Uninterrupted Power Supply (UPS). Do not connect the system to un-stabilized power supply. Instrument damaged due to fluctuations in power supply will not be covered under warranty.
- 10.1.2. Continuous compressed air supply for Purge air with pressures (3 bar to 5 bar) is necessary.
- 10.1.3. Do not switch ON the instrument without purge air supply.
- 10.1.4. Do keep the instrument free from dust, vibrations, heat, and electromagnetic noise generating sources.
- 10.1.5. Do keep one set of airline filters and regulators for purge air supply right before supplying to the instrument.
- 10.1.6. Do provide Weather shield cover for transmitter and receiver units.
- 10.1.7. Do keep the monitor unit in a room environment. Ensure the room temperature do not increase above 35 °C.
- 10.1.8. Do not try to open/repair/service the instrument in case of faulty measurements. Please contact customer support.
- 10.1.9. Do not look directly at the laser. It may cause serious eye injury, loss of vision, etc.
- 10.1.10. Do not touch the electrical terminals with bare hands.
- 10.1.11. The TFT/LCD screens may be damaged by sharp objects. Do not use any stylus, metal, wood objects to operate display screen.

10.2. PERIODIC MAINTENANCE

- 10.2.1. Daily Maintenance: Check the compressed air supply for minimum 3 bar air supply. Check the measurement values for usual reading.
- 10.2.2. Monthly Maintenance: Check the laser light by inserting a white sheet of paper. The laser light dot must be within 6-7mm in diameter. Clean the transmitter and receiver optical components with extreme care.
- 10.2.3. Quarterly Maintenance: Check the connection terminals for loose connection. If any disconnection/loose-connection observed, fasten the screws on the terminals to make

a close-fitting connection.

10.2.4. Yearly Maintenance: Check whether the range of the stack has changed. Recalibrate the instrument. Check for wear and tear in the connecting cables.

10.3. TROUBLESHOOTING

Some troubleshooting methods are listed below.

S. No	Problem	Possible Causes	Troubleshooting
1.	Power LED is OFF	1. Power supply failure 2. SMPS failure	1. Contact the plant electrical engineer to check the power supply to the system 2. Replace SMPS
3.	Power LED is ON Status LED is permanently OFF	Motherboard failure	Please contact customer support
4.	Power LED-ON Status LED-blinking Measurement value –zero always	Calibration error	Perform Recalibration
5.	Status LED-blinking Power LED-ON Full-scale measurement always	1. Receiver unit Wire disconnected 2. Laser unit malfunction	1. Make the connection 2. Contact customer support
6.	Measurement values higher than usual	1. Accumulation of dust on the transmitter and receiver optical components 2. Laser light out of focus 3. Transmitter/Receiver unit malfunction	1. Contact a trained person to clean the transmitter and receiver optical components. Check air supply. 2. Contact customer support 3. Contact customer support
7.	Measurement values lower than usual	1. Dust concentration very low. 2. Lower particle size	Perform Manual reading & Recalibration
8.	4-20mA Current -values not matching the measured values	1. Cable damaged 2. Monitor unit malfunction	1. Check the cable connection 2. Contact customer support